

Hive

Sure! Let's dive into the concept of Hive and its architecture in simple terms.

Hive: A Data Warehouse for Big Data

Hive is like a big library for data, specifically designed to help us manage and analyze large amounts of information. Imagine you have a huge collection of books (data) from different sources, and you want to find specific information quickly. Hive helps you organize these books so you can easily read and understand them. It uses a language similar to SQL, which is like a common language for talking to databases, making it easier for people who already know SQL to learn how to use Hive.

The Architecture of Hive

Think of Hive's architecture as a well-organized team working together to help you find the information you need. There are three main parts:

1.

Hive Client: This is like the librarian who helps you request books. It connects different applications to Hive.

2.

Hive Services: This is where the magic happens! It processes your requests and runs the queries you ask, just like a librarian fetching the right books for you.

3.

Hive Storage and Computing: This is the storage room where all the books (data) are kept. It ensures that everything is stored safely and can be accessed quickly.

Hive

Hive is data warehouse software within Hadoop that is designed for reading, writing and managing tabular-type datasets and data analysis:

- It is scalable, fast, and easy to use
- Hive Query Language (HiveQL) is inspired by SQL, making it easier for users to grasp concepts
- It supports data cleansing and filtering depending on users' requirements



A data warehouse stores historical data from many different sources so that you can analyze and extract insights from it

- Hive is a data warehouse software within Hadoop that is designed to read, write, and manage large and tabular-type datasets and data analysis. A data warehouse stores historical data from many different sources so that you can analyze and extract insights from it.
- These insights are used for reporting. Hive is scalable and fast because it is designed to work on petabytes of data. It is easy to use if you are familiar with SQL because Hive query language, or Hive QL, is based on SQL.
- This makes learning quicker because users who already deal with relational databases can easily grasp the concepts. Hive supports the following file formats: Sequence files consisting of binary key value pairs, record columnar files that store columns of a table in a columnar database, and text or flat files. Hive allows for data cleaning and filtering tasks according to users' requirements.

Hive and traditional RDBMS compared



Hive and traditional RDBMS compared

Traditional RDBMS	Hive
Used to maintain a Database and uses SQL	Used to maintain a data warehouse using Hive query language
Suited for real-time/dynamic data analysis like data from sensors	Suited for static data analysis like a text file containing names
Designed to read and write as many times as it needs	Designed on the methodology of write once, read many
Maximum data size it can handle is terabytes	Maximum data size it can handle is petabytes
Enforces that the schema must verify loading data before it can proceed	Doesn't enforce the schema to verify loading data
May not always have built-in for support data partitioning	Supports partitioning



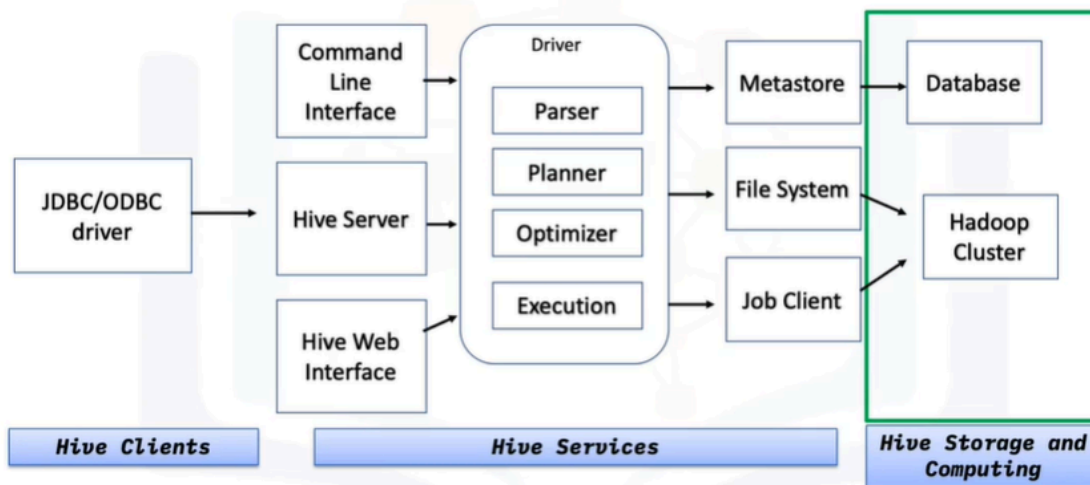
Partitioning means dividing the table into parts based on the values of a particular column, such as date or city

- RDBMS is a type of database management system specifically designed for relational databases. RDBMS is the acronym for Relational Database Management System. A relational database is a database that stores data in a structured format using rows and columns known as a table.
- Let's look at the differences between a traditional RDBMS and Hive. Traditional RDBMS is used to maintain a database and uses the structured query language known as SQL. Hive is used to maintain a data warehouse and uses the Hive query language, a language inspired by SQL. Traditional RDBMS is

suited for real-time data analysis, like data from sensors. Hive is suited for static data analysis.

- Traditional RDBMS allows for as many read and write operations a user needs. Hive is based on the write once, read many methodology. Traditional RDBMS can handle up to terabytes of data. Hive is designed to handle as much as petabytes of data. Another difference is that traditional RDBMS enforces that the schema must verify loading data before it can proceed.
- Hive doesn't enforce the schema to verify loading data. Lastly, traditional RDBMS may not always have built-in for support data partitioning. Hive supports partitioning. Partitioning means dividing the table into parts based on the values of a particular column such as date or city.

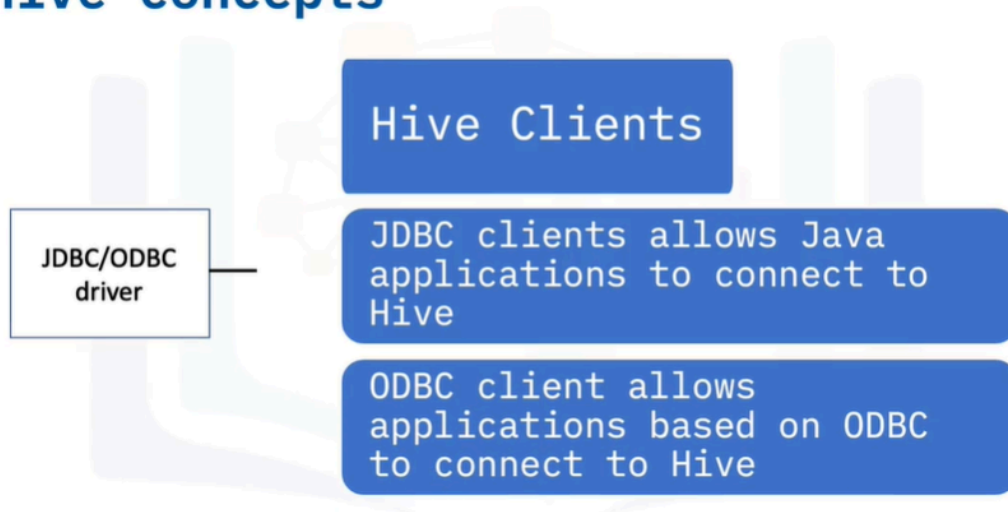
Hive architecture



- This is what the Hive architecture looks like.
- There are three main parts of the architecture: The Hive client: Hive provides different drivers for communication depending on the type of application. For example, for Java based applications, it uses JDBC drivers, and other types of applications will use ODBC drivers. These drivers communicate with the servers.

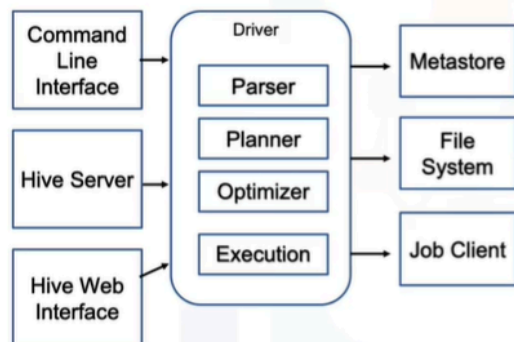
- Hive services: Client interactions are done through the Hive services. Any query operations are done here.
- The command line interface acts as an interface for the Hive service. The driver takes in query statements, monitors the progress and life cycle of each session, and stores metadata generated from the query statements.
- The meta store stores the metadata, the data and information about each table, such as the location and schema. The meta store, file system, and job client in turn communicate with Hive storage and computing to perform the following: Metadata information of tables are stored in some sort of database and query results and data loaded in the tables are stored in Hadoop cluster or HDFS.

Hive concepts



- Let's look at the concepts in detail. Components in the Hive clients include: The JDBC client, which allows Java-based applications to connect to Hive, and the ODBC client, which allows applications based on the ODBC protocol to connect to Hive.

Hive concepts



Hive Services

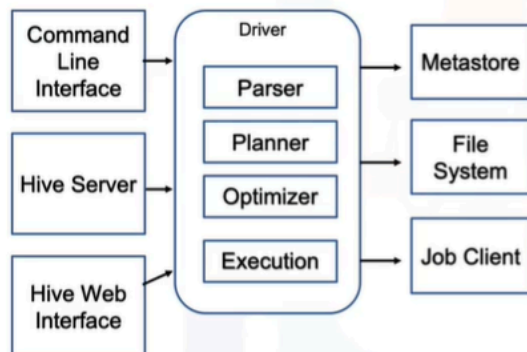
Hive Server to enable queries

The driver receives query statements

The optimizer is used to split tasks efficiently

The executor executes tasks after the optimizer

Hive concepts



Hive Services

Metastore stores the metadata information about the tables

The driver receives query statements

The optimizer is used to split tasks efficiently

The executor executes tasks after the optimizer

- The Hive services section of the architecture is in charge of performing queries. The Hive server is used to run queries and allows multiple clients to submit requests. It is built to support the JDBC and ODBC clients. The driver receives query statements submitted through the command line and sends the query to the compiler after initiating a session. The optimizer performs transformations on the execution plan and splits the tasks to help speed up

and improve efficiency. The executor executes tasks after the optimizer has split the tasks. The meta store is storage for the metadata, which is information about the table. The meta store is in charge of keeping this information in a central place.

Summary

In this video, you learned:

- Hive is a data warehouse software for reading, writing, and managing datasets
- Although Hive works very similarly to traditional RDBMS, they are slightly different
- The three main parts of the Hive architecture are Hive Client, Hive Services, and Hive Storage and Computing